

TIC 235512374

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.744060 d
Epoch (t0) 1472.0090 BTJD
Duration 0.0840 d
Depth 3.410 % (34,100 ppm)
SNR 173.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

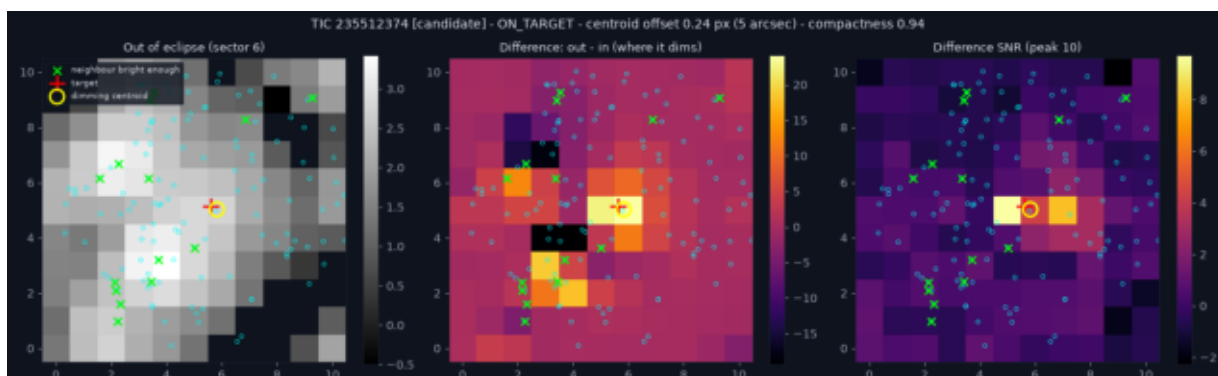
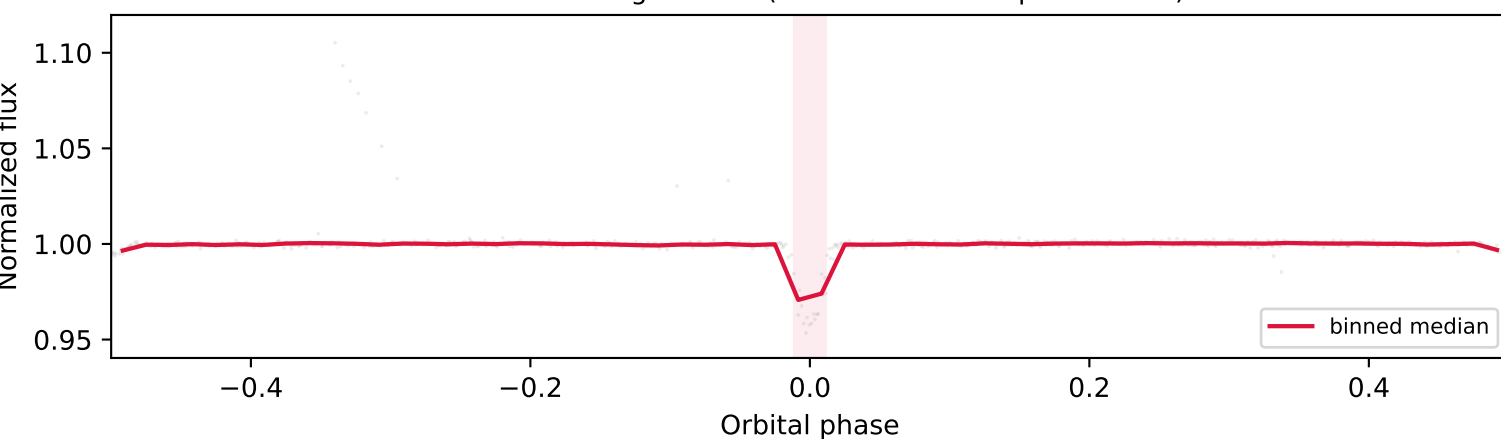
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 132.8 (gate ≥ 7.0)
Odd/even z-score 1.79 (consistent)
Secondary eclipse detected, phase -0.49, 6.6 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 119.5 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 9.71 (gate ≥ 5.0) Compactness 0.94 (gate ≥ 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 235512361
1.60 px away, dTmag +2.72 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.